

Terraform assignment

```
[ec2-user@ip-10-0-1-131 ~]$ sudo yum-config-manager --add-repo https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
Adding repo from: https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
[ec2-user@ip-10-0-1-131 ~]$ sudo yum -y install terraform
Hashicorp Stable - x86_64 12 MB/s | 1.3 MB 00:00
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Installing:
terraform x86_64 1.7.1-1 hashicorp 26 M
Installing dependencies:
git x86_64 2.40.1-1.amzn2023.0.1 amazonlinux 57 k
git-core x86_64 2.40.1-1.amzn2023.0.1 amazonlinux 4.3 M
git-core-doc noarch 2.40.1-1.amzn2023.0.1 amazonlinux 2.6 M
perl-Error noarch 1:0.17029-5.amzn2023.0.2 amazonlinux 41 k
perl-File-Find noarch 1.37-477.amzn2023.0.6 amazonlinux 26 k
perl-Git noarch 2.40.1-1.amzn2023.0.1 amazonlinux 45 k
perl-TermReadKey x86_64 2.38-9.amzn2023.0.2 amazonlinux 36 k
perl-lib x86_64 0.65-477.amzn2023.0.6 amazonlinux 15 k
=====
Transaction Summary
=====
Install 9 Packages

Total download size: 33 M
Installed size: 114 M
Downloading Packages:
(1/9): git-2.40.1-1.amzn2023.0.1.x86_64.rpm 830 kB/s | 57 kB 00:00
```

[IAM](#) > [Security credentials](#) > Create access key

Step 1
[Alternatives to root user access](#)
[keys](#)

Step 2
Retrieve access key

Retrieve access key [Info](#)

Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
AKIA4SEGK4ISH2R5CI5F	***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#) [Done](#)

```
Complete!
[ec2-user@ip-10-0-1-131 ~]$ export AWS_ACCESS_KEY_ID = AKIA4SEGK4ISH2R5CI5F
-bash: export: '=': not a valid identifier
[ec2-user@ip-10-0-1-131 ~]$ export AWS_ACCESS_KEY_ID=AKIA4SEGK4ISH2R5CI5F
[ec2-user@ip-10-0-1-131 ~]$ export AWS_SECRET_ACCESS_KEY=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
[ec2-user@ip-10-0-1-131 ~]$ sudo yum install -y aws-cli
Last metadata expiration check: 0:04:48 ago on Sat Jan 27 06:00:08 2024.
Package awscli-2.14.5-1.amzn2023.0.1.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-131 ~]$ aws --version
aws-cli/2.14.5 Python/3.9.16 Linux/6.1.72-96.166.amzn2023.x86_64 source/x86_64.amzn.2023 prompt/off
[ec2-user@ip-10-0-1-131 ~]$ |
[ec2-user@ip-10-0-1-131 tf_bundle]$
[ec2-user@ip-10-0-1-131 tf_bundle]$ ll
total 16
-rw-rw-r--. 1 ec2-user ec2-user 315 Jan 27 06:28 ec2.tf
-rw-rw-r--. 1 ec2-user ec2-user 1344 Jan 27 06:28 network.tf
-rw-rw-r--. 1 ec2-user ec2-user 44 Jan 27 06:28 provider.tf
-rw-rw-r--. 1 ec2-user ec2-user 561 Jan 27 06:28 vpc.tf
[ec2-user@ip-10-0-1-131 tf_bundle]$ |
```

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ terraform init
```

Initializing the backend...

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.34.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ |
```

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ aws configure
AWS Access Key ID [None]: AKIA4SEGK4ISE7GIGH73
AWS Secret Access Key [None]: vWB1JTvpUKc3/TqOy+T3cakKrV1Mv2DhJNcYICp
Default region name [None]: ap-south-1
Default output format [None]:
```

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_instance.jayas-tf-web1 will be created
+ resource "aws_instance" "jayas-tf-web1" {
  + ami              = "ami-0455d910dab418e8f"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + cpu_core_count   = (known after apply)
  + cpu_threads_per_core = (known after apply)
  + disable_api_stop  = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized     = (known after apply)
  + get_password_data = false
  + host_id           = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile = (known after apply)
  + id                = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle = (known after apply)
  + instance_state     = (known after apply)
  + instance_type       = "t2.micro"
  + ipv6_address_count  = (known after apply)
  + ipv6_addresses      = (known after apply)
  + key_name            = "ec2_key"
```

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ terraform validate
```

Success! The configuration is valid.

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ terraform plan -out terraform.out
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_instance.jayas-tf-web1 will be created
+ resource "aws_instance" "jayas-tf-web1" {
  + ami              = "ami-0455d910dab418e8f"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + cpu_core_count   = (known after apply)
  + cpu_threads_per_core = (known after apply)
  + disable_api_stop  = (known after apply)
  + disable_api_termination = (known after apply)
```

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.jayas-tf-web1 will be created
  + resource "aws_instance" "jayas-tf-web1" {
    + ami                    = "ami-0455d910dab418e8f"
    + arn                   = (known after apply)
    + associate_public_ip_address = (known after apply)
    + availability_zone       = (known after apply)
    + cpu_core_count         = (known after apply)
    + cpu_threads_per_core   = (known after apply)
    + disable_api_stop       = (known after apply)
    + disable_api_termination = (known after apply)
    + ebs_optimized          = (known after apply)
    + get_password_data      = false
    + host_id                = (known after apply)
    + host_resource_group_arn = (known after apply)
    + iam_instance_profile   = (known after apply)
    + id                     = (known after apply)

Plan: 7 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_vpc.jayas-tf-vpc: Creating...
aws_vpc.jayas-tf-vpc: Still creating... [10s elapsed]
aws_vpc.jayas-tf-vpc: Creation complete after 11s [id=vpc-06f94cabba413cec1]
aws_security_group.ssh-allowed: Creating...
aws_internet_gateway.jayas-tf-igw: Creating...
aws_subnet.jayas-tf-subnet-public-1: Creating...
aws_internet_gateway.jayas-tf-igw: Creation complete after 1s [id=igw-01f8ac48588ce0c45]
aws_route_table.jayas-tf-public-crt: Creating...
aws_route_table.jayas-tf-public-crt: Creation complete after 0s [id=rtb-0c4ac0d66b159e1d]
aws_security_group.ssh-allowed: Creation complete after 2s [id=sg-0d5af8b5b13c97aed]
aws_subnet.jayas-tf-subnet-public-1: Still creating... [10s elapsed]
aws_subnet.jayas-tf-subnet-public-1: Creation complete after 11s [id=subnet-0108a4e0a5cdf767e]
aws_instance.jayas-tf-web1: Creating...
aws_route_table_association.jayas-tf-crt-public-subnet-1: Creating...
aws_route_table_association.jayas-tf-crt-public-subnet-1: Creation complete after 0s [id=rtbassoc-007568b5f2c024e27]
aws_instance.jayas-tf-web1: Still creating... [10s elapsed]
aws_instance.jayas-tf-web1: Still creating... [20s elapsed]
aws_instance.jayas-tf-web1: Still creating... [30s elapsed]
aws_instance.jayas-tf-web1: Creation complete after 31s [id=i-08f4e4d51f56d5b17]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
```

```
[ec2-user@ip-10-0-1-131 tf_bundle]$ terraform apply terraform.out

aws_vpc.jayas-tf-vpc: Creating...
aws_vpc.jayas-tf-vpc: Still creating... [10s elapsed]
aws_vpc.jayas-tf-vpc: Creation complete after 11s [id=vpc-0dc6d0e34749c5081]
aws_internet_gateway.jayas-tf-igw: Creating...
aws_subnet.jayas-tf-subnet-public-1: Creating...
aws_security_group.ssh-allowed: Creating...
aws_internet_gateway.jayas-tf-igw: Creation complete after 1s [id=igw-0f1eac6f0df58522]
aws_route_table.jayas-tf-public-crt: Creating...
aws_route_table.jayas-tf-public-crt: Creation complete after 0s [id=rtb-0780a2f81d1e1c32f]
aws_security_group.ssh-allowed: Creation complete after 2s [id=sg-0ad5eb87d07485f90]
aws_subnet.jayas-tf-subnet-public-1: Still creating... [10s elapsed]
aws_subnet.jayas-tf-subnet-public-1: Creation complete after 11s [id=subnet-0bcff9bccf02fb5ca]
aws_instance.jayas-tf-web1: Creating...
aws_route_table_association.jayas-tf-crt-public-subnet-1: Creating...
aws_route_table_association.jayas-tf-crt-public-subnet-1: Creation complete after 0s [id=rtbassoc-0771af56a1934076f]
aws_instance.jayas-tf-web1: Still creating... [10s elapsed]
aws_instance.jayas-tf-web1: Still creating... [20s elapsed]
aws_instance.jayas-tf-web1: Still creating... [30s elapsed]
aws_instance.jayas-tf-web1: Creation complete after 31s [id=i-09f642cd0a9b026a2]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
[ec2-user@ip-10-0-1-131 tf_bundle]$
```

☒
i-08f4e4d51f56d5b17
Running
t2.micro
Initializing
View alarms
ap-south-1a
ec2-13-12

Instance: i-08f4e4d51f56d5b17

←↻⚠ Not secure | 13.127.253.253

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](#). Commercial support is available at [nginx.com](#).

Thank you for using nginx.

🔍 Search

☐	Name	VPC ID	State
☐	terra_route_2	vpc-06f94cabba413cec1	✔ Available
☐	-	vpc-05bc5283de6e92427	✔ Available
☐	terra_route_1	vpc-05a4c6eb701afabe8	✔ Available

☐ subnet-0108a4e0a5cdf767e

✔ Available

[vpc-06f94cabba413cec1](#) | jayas...

10.2.1.0/24

subnet-02f571eef6f4e0d5f

Details

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

☐	terra_route_2	rtb-0c4acf0d66b159e1d	subnet-0108a4e0a5cdf7...	-	No	vpc-06f94cabba413cec1
☐	terra-tf-2	rtb-0e759a9ba8cc8023f	2 subnets	-	No	vpc-05a4c6eb701afabe8

Select a route table

Internet gateways (3) Info

🔄 Actions ▼ Create internet gateway

🔍 Search

< 1 > ⚙

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	terra_route	igw-01f8ac48588ce0c45	✔ Attached	vpc-06f94cabba413cec1 jayas-tf-vpc	863570158116
☐	terra_route_2	igw-0b86a0d069996a4e6	✔ Attached	vpc-05a4c6eb701afabe8 jayas-vpc	863570158116
☐	terra_route_public	igw-0e83132dbafe9a167	✔ Attached	vpc-05bc5283de6e92427	863570158116